Program : **Diploma in Instrumentation Engineering**	
Course Code : **3089**	Course Title: **Product Design and development workshop**
Semester : **3**	Credits: **No Credit**
Course Category: **Program Core**	
Periods per week: **4 (L:0 T:0 P:4)**	Periods per semester: **60**

Course Objectives:

- To create the skill that is required to develop simple hobby projects using electronic and instrument components, Relays, Arduino, Raspberry pi and compatibles.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic concepts of computer programming.		Introduction to IT systems	1
Concepts of electrical and electronic components		Fundamentals of Electrical & Electronics Lab	2
Concepts and working of electrical and electronic instruments		Electronic Instrumentation lab	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Construct simple models and hobby circuits	16	Applying
CO2	Develop simple interfacing applications and products using Arduino.	15	Applying

CO3	Develop simple interfacing applications and products using Raspberry Pi.	13	Applying
CO4	Set up various circuits using relay logic	13	Applying
	Lab Test	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1			3				
CO2			3	3			
CO3			3	3			
CO4			3				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Construct simple models and hobby circuits by soldering.**		
M1.1	Identify tools and consumables used for soldering and de-soldering of "through hole" PCBs – soldering iron of different wattage, temperature-controlled soldering station, soldering iron stand, fume extractor, solder of various grade, flux, nipper, wire stripper, needle node plier, tweezer, de-soldering pump, de-soldering station, single layer, multi-layer, through hole and SMD PCBs, etc.	1	Applying
M1.2	Identify tools used to perform soldering and de-soldering of SMD PCBs – soldering station, electronic rework station etc.	1	Applying
M1.3	Familiarize safety precautions in handling soldering tools – personal safety, safety of components and PCBs, ESD safety.	1	Applying

M1.4	Model 2x2 mesh by soldering single strand copper wire. (Perform tinning before soldering)	1	Applying
M1.5	Model 2x2x2 mesh polygonal cube by soldering single strand copper wire. (Perform tinning before soldering)	2	Applying
M1.6	Construct hobby circuits using knock down electronic kits containing pre-fabricated PCBs– LED bulbs, LED running displays, serial lights, Xmas lights, power supply. (Assemble any one hobby electronic PCB. The list shown is only indicative)	6	Applying
M1.7	Construct hobby circuits using general purpose PCB (dot board). (Assemble any one simple hobby electronic circuit)	4	Applying
CO2	**Develop simple interfacing applications using Arduino.**		
M2.1	Familiarize Arduino UNO, Arduino IDE	1	Understanding
M2.2	Blink LED connected to P13 at 1 sec interval. Interface relay module with port and switch relay on and off at different intervals.	1	Applying
M2.3	Connect PIR sensor (motion sensor) to port and switch relay connected to port on detecting motion.	1	Applying
M2.4	Familiarize DC motor control shield and control DC motor for forward movement, reverse movement and stop.	1	Applying
M2.5	Use wheeled robot kit (containing two geared DC motors, caster wheel, Arduino Uno, motor driver shield, 9V battery and line sensor) to make line follower robot.	2	Applying
M2.6	Familiarize blue tooth module (HC 05 or HC 06) to switch relay connected to port using smart phone.	1	Applying
M2.7	Use wheeled robot kit and blue tooth module to control robot using smart phone.	2	Applying
M2.8	Use LCD shield and display your name and class number.	2	Applying

M2.9	Implement suitable Do it Yourself (DIY) project from internet and implement using Arduino and compatibles. (Open ended experiment)	2	Applying
M2.10	Develop a product using Arduino	3	Applying
	Lab Test 1	1.5	
CO3	**Develop simple interfacing applications using Raspberry pi.**		
M3.1	Familiarize Raspberry Pi and compatibles.	1.5	Understanding
M3.2	Install suitable raspberry pi operating system	1.5	Applying
M3.3	Familiarize raspberry pi operating system.	2	Applying
M3.4	Interface raspberry pi IO ports for input and output.	2	Applying
M3.5	Implement suitable DIY project from internet using Raspberry pi (Open ended experiment)	2	Applying
M3.6	Develop a product using Raspberry pi	4	Applying
CO4	**Set up various circuits using relay logic**		
M4.1	Familiarize different types of Relays SPDT, DPDT, contactor, high current relays, on delay, off delay timer, NO, NC push button, solid state relays etc, NO NC contacts and ladder diagram	2	Applying
M4.2	Realization of logic gates using relays.	2	Applying
M4.3	Wiring of relay latch circuit.	2	Applying
M4.4	water level controller circuit using 2 relays and 2 float switches	2	Applying
M4.5	Wire a circuit to switch on a device after 10 sec delays using on delay timer.	2	Applying
M4.6	Wire a circuit to switch off a lamp after 10 seconds using off delay timer. (switch on. light should be on Switch off continue glowing10 seconds then off)	2	Applying
M4.7	Operate a 230 V AC lamp using a dc (24v) relay.	1	Applying
	Lab Test 2	1.5	

Text /Reference:

T/R	Book Title/Author
R1	Jonathan Oxer, Hugh Blemings, Practical Arduino: cool projects for open source hardware, Apress; Distributed by Springer-Verlag, Year: 2009
R2	Massimo Banzi, Getting Started with Arduino, Make: Projects, Make, Year: 2008
R3	Geoffrey Boothroyd, Winston Knight, Peter Dewhurst, Product Design for Manufacture and Assembly, M. Dekker, Year: 2002

Online Resources:

Sl.No	Website Link
1	https://www.arduino.cc/
2	https://www.raspberrypi.org/